

YEAR 7 MATHEMATICS

Chapter 6

Geometrical figures

WORKSHEETS

6

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

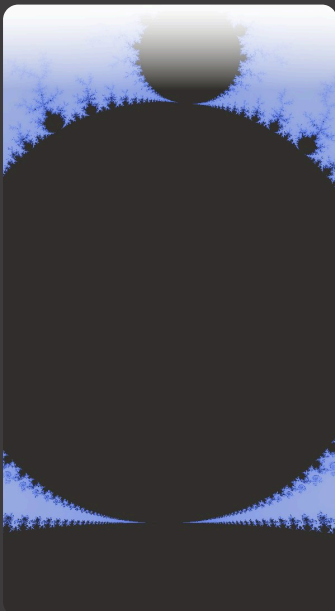
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
6.01	Transforming shapes	3		☐
6.02	Combined transformations	4		☐
6.03	Line symmetry	5		☐
6.04	Rotational symmetry	6		☐
6.05	Classifying triangles	7		☐
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In this chapter you will

- reflect, translate and rotate shapes
- find line and rotational symmetry
- classify triangles and quadrilaterals
- use the angle sums of triangles and quadrilaterals
- use the exterior angle of a triangle
- describe the properties of special quadrilaterals



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the top bulb of the set and the lacy edge around it. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–18

Translate (slide) the point.
Right and up add; left and down subtract.



WE DO With your teacher: copy the working.

E1 (5, 2): 2 right, 4 down

E2 (–2, 3): 6 left, 3 down

YOU DO Your turn.

1 (–6, –1): 6 right, 3 up **(0, 2)**

2 (3, 6): 4 right, 5 up

3 (–2, 3): 3 left, 1 down

4 (–5, –3): 4 left, 6 down

5 (3, 0): 2 right, 5 down

6 (–4, 3): 4 right, 1 down

7 (5, –1): 1 left, 4 up

8 (–3, 5): 3 left, 1 up

9 (–6, 6): 1 left, 4 up

10 (3, –3): 2 right, 1 down

11 (3, –4): 1 right, 5 down

12 (3, 2): 1 left, 5 down

13 (6, 2): 6 right, 1 down

14 (5, 6): 3 right, 6 down

15 (–4, –5): 1 right, 5 up

16 (0, –6): 6 right, 2 up

17 (–6, 2): 3 left, 3 down

18 (–5, 1): 3 right, 4 up

Medium

Questions 19–28

Reflect the point in the axis.
In the x-axis, change the sign of y; in the y-axis, change the sign of x.

WE DO With your teacher: copy the working.

E1 (–8, 5) in the x-axis

E2 (–8, –6) in the x-axis

YOU DO Your turn.

19 (8, 5) in the x-axis

change the sign of y

→ **(8, –5)**

20 (–2, 3) in the y-axis

change the sign of x

→ **(____, ____)**

21 (–8, –5) in the x-axis

→ **(____, ____)**

22 (–5, 3) in the y-axis

23 (–6, 7) in the x-axis

24 (–3, –5) in the x-axis

25 (5, –6) in the x-axis

26 (–7, 7) in the x-axis

27 (–6, 2) in the x-axis

28 (–6, 8) in the y-axis

Challenging

Questions 29–35

Rotate the point about the origin O. Use the rule, then check on the number plane.



WE DO With your teacher: copy the working.

E1 Rotate (2, –2) by 180° about O.

E2 Rotate (3, –4) 90° clockwise about O.

E3 Rotate (–4, 6) 90° anticlockwise about O.

YOU DO Your turn.

29 Rotate (–5, –6) by 180° about O.

180°: (x, y) → (–x, –y)

(____, ____) → (____, ____)

30 Rotate (–2, –4) 90° clockwise about O.

____° clockwise: (x, y) → (y, –x)

(____, ____) → (____, ____)

31 Rotate (5, 6) 90° anticlockwise about O.

32 Rotate (–6, 3) by 180° about O.

33 Rotate (–2, 6) 90° clockwise about O.

34 Rotate (–5, 6) 90° anticlockwise about O.

35 Rotate (3, 1) by 180° about O.

Easy

Questions 1–13

Name the transformation: translation, reflection or rotation.

WE DO With your teacher: copy the working.

E1 a slide

E2 a flip

YOU DO Your turn.

1 a turn

rotation

2 a mirror image

3 moving 5 units up

4 a clock hand moving

5 your reflection in a lake

6 a lift going up

7 a wheel turning

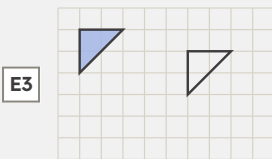
8 a door opening

9 a car driving straight

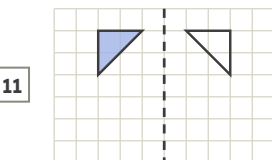
10 a butterfly's wings

Name the transformation that moves the shaded shape to the outlined shape.

WE DO With your teacher: copy the working.

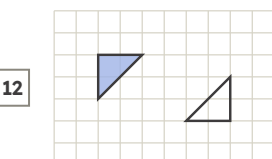


YOU DO Your turn.

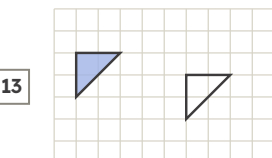


11

reflection



12



13

Medium

Questions 14–18

Do both translations. Add the moves in each direction, then give one single translation.

WE DO With your teacher: copy the working.

E1 7 right, 6 up; then 6 left, 2 down

E2 1 right, 6 up; then 6 left, 2 down

YOU DO Your turn.

14 2 right, 5 up; then 6 left, 3 down

across: $2 - 6 = -4$

up/down: $5 - 3 = 2$

4 left, 2 up

15 5 right, 4 up; then 2 left, 1 down

across: $5 - 2 = 3$

up/down: $4 - 1 = 3$

___ right, ___ up

16 7 right, 4 up; then 6 left, 5 down

across: ___ - ___ = ___

up/down: ___ - ___ = ___

___ right, ___ down

17 7 right, 4 up; then 5 left, 3 down

18 1 right, 4 up; then 2 left, 7 down

Challenging

Questions 19–26

Do the transformations in order. Write where the point is after each one.

WE DO With your teacher: copy the working.

E1 $(-2, -3)$: 4 right, then reflect in the x-axis

E2 $(4, 2)$: rotate 180° about O, then 5 left

YOU DO Your turn.

19 $(-5, 4)$: 2 right, then reflect in the x-axis

2 right: $(-3, 4)$

reflect in x-axis: $(_, _)$

20 $(-2, 4)$: rotate 180° about O, then 4 left

rotate 180° : $(_, _)$

___ left: $(_, _)$

21 $(1, -4)$: 3 right, then reflect in the x-axis

22 $(2, 1)$: rotate 180° about O, then 4 left

23 $(-4, 3)$: 4 right, then reflect in the x-axis

24 $(-3, 3)$: rotate 180° about O, then 1 left

25 $(-4, 3)$: 5 right, then reflect in the x-axis

26 $(-1, -3)$: rotate 180° about O, then 1 left

Easy

Questions 1–9

How many axes (lines) of symmetry does the shape have? Fold it in your head.

WE DO With your teacher: copy the working.

E1

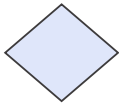


E2



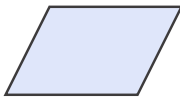
YOU DO Your turn.

1

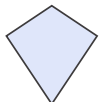


2

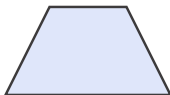
2



3



4



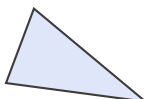
5



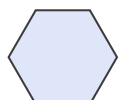
6



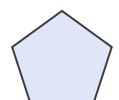
7



8



9



Medium

Questions 10–19

How many axes of symmetry does a regular polygon have? It has one for every side.

WE DO With your teacher: copy the working.

E1 a regular polygon with 16 sides

E2 a regular polygon with 3 sides

YOU DO Your turn.

10 a regular polygon with 13 sides

13 sides → 13 axes

answer: 13

11 a regular polygon with 7 sides

7 sides → 7 axes

answer: ____

12 a regular polygon with 12 sides

____ sides → ____ axes

answer: ____

13 a regular polygon with 20 sides

14 a regular polygon with 19 sides

15 a regular polygon with 15 sides

16 a regular polygon with 10 sides

17 a regular polygon with 4 sides

18 a regular polygon with 18 sides

19 a regular polygon with 9 sides

Challenging

Questions 20–22

Name the shape, then draw its axes of symmetry on it. How many are there?

WE DO With your teacher: copy the working.

E1 Name it and draw the axes.

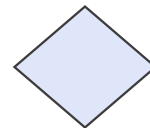


E2 Name it and draw the axes.



YOU DO Your turn.

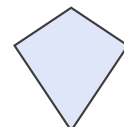
20 Name it and draw the axes.



shape: rhombus

axes of symmetry: ____

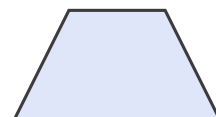
21 Name it and draw the axes.



shape: ____

axes of symmetry: ____

22 Name it and draw the axes.



6.04 Rotational symmetry

Easy

Questions 1–9

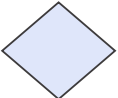
What is the order of rotational symmetry? (How many times does it fit onto itself in one turn? Write 1 for none.)

WE DO With your teacher: copy the working.

E1  _____

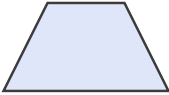
E2  _____

YOU DO Your turn.

1  _____ **2** _____

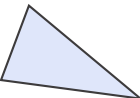
2  _____

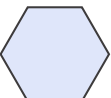
3  _____

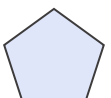
4  _____

5  _____

6  _____

7  _____

8  _____

9  _____

Medium

Questions 10–21

A regular polygon: find the order of rotational symmetry, then the smallest angle of turn ($360^\circ \div \text{order}$).

WE DO With your teacher: copy the working.

E1 24 sides **E2** 15 sides

YOU DO Your turn.

10 5 sides **11** 36 sides

order 5 order 36

$360^\circ \div 5 = 72^\circ$ $360^\circ \div \text{ } = \text{ }^\circ$

12 10 sides **13** 30 sides

order _____

$360^\circ \div \text{ } = \text{ }^\circ$

14 18 sides **15** 3 sides

16 8 sides **17** 20 sides

18 6 sides **19** 4 sides

20 9 sides **21** 12 sides

Challenging

Questions 22–30

Compare the two kinds of symmetry for each shape.

WE DO With your teacher: copy the working.

E1 a square

E2 a rectangle

YOU DO Your turn.

22 a rhombus

axes of symmetry: 2

order of rotation: _____

23 a parallelogram

axes of symmetry: _____

order of rotation: _____

24 a kite

25 an isosceles trapezium

26 an equilateral triangle

27 an isosceles triangle

28 a scalene triangle

29 a regular hexagon

30 a regular pentagon

6.05 Classifying triangles

Easy

Questions 1–24

Classify the triangle by its angles: acute, right or obtuse.

WE DO With your teacher: copy the working.

E1 $40^\circ, 85^\circ, 55^\circ$

E2 $50^\circ, 90^\circ, 40^\circ$

YOU DO Your turn.

1 $45^\circ, 45^\circ, 90^\circ$

right

2 $30^\circ, 70^\circ, 80^\circ$

3 $65^\circ, 90^\circ, 25^\circ$

4 $25^\circ, 70^\circ, 85^\circ$

5 $30^\circ, 75^\circ, 75^\circ$

6 $60^\circ, 35^\circ, 85^\circ$

7 $75^\circ, 90^\circ, 15^\circ$

8 $115^\circ, 50^\circ, 15^\circ$

9 $120^\circ, 20^\circ, 40^\circ$

10 $75^\circ, 25^\circ, 80^\circ$

11 $50^\circ, 25^\circ, 105^\circ$

12 $100^\circ, 20^\circ, 60^\circ$

Classify the triangle by its sides: equilateral, isosceles or scalene.

WE DO With your teacher: copy the working.

E3 3 cm, 3 cm, 3 cm

YOU DO Your turn.

13 6 cm, 7 cm, 10 cm

scalene

14 12 cm, 12 cm, 13 cm

15 11 cm, 11 cm, 2 cm

16 10 cm, 10 cm, 10 cm

17 11 cm, 11 cm, 10 cm

18 9 cm, 9 cm, 9 cm

19 10 cm, 11 cm, 14 cm

20 5 cm, 8 cm, 10 cm

21 4 cm, 4 cm, 4 cm

22 10 cm, 10 cm, 5 cm

23 6 cm, 9 cm, 11 cm

24 8 cm, 8 cm, 8 cm

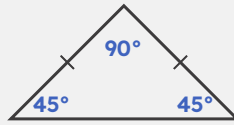
Medium

Questions 25–27

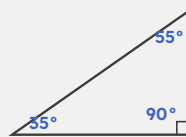
Classify the triangle by its angles and by its sides (equal marks show equal sides).

WE DO With your teacher: copy the working.

E1



E2



YOU DO Your turn.

25



angles: obtuse

sides: scalene

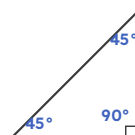
26



angles: obtuse

sides: _____

27



angles: _____

sides: _____

Challenging

Questions 28–33

Can the triangle be drawn? Write yes or no, and explain.

WE DO With your teacher: copy the working.

E1 a right-angled isosceles triangle

E2 a triangle with two right angles

YOU DO Your turn.

28 an obtuse equilateral triangle

no

because: equilateral angles are all 60°

29 a right-angled scalene triangle

because: e.g. $90^\circ, 30^\circ, 60^\circ$

30 a triangle with two obtuse angles

31 an obtuse isosceles triangle

32 an acute scalene triangle

33 a right-angled equilateral triangle

6.06 Angle sum of a triangle

Easy

Questions 1–30

The angles of a triangle add to 180° . Find the third angle.



WE DO With your teacher: copy the working.

E1 31° and 32°

E2 101° and 44°

YOU DO Your turn.

1 18° and 64°

98°

2 47° and 73°

3 48° and 34°

4 68° and 69°

5 35° and 23°

6 94° and 47°

7 25° and 39°

8 66° and 89°

9 61° and 42°

10 50° and 60°

11 86° and 64°

12 70° and 78°

13 97° and 37°

14 68° and 33°

15 41° and 106°

16 106° and 51°

17 58° and 23°

18 68° and 48°

19 68° and 18°

20 91° and 29°

21 98° and 43°

22 111° and 43°

23 71° and 31°

24 84° and 55°

25 61° and 54°

26 76° and 29°

27 45° and 33°

28 16° and 124°

29 80° and 69°

30 105° and 34°

Medium

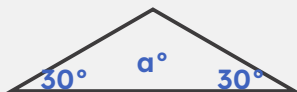
Questions 31–33



Find a . Write the equation first.

WE DO With your teacher: copy the working.

E1 Find a .

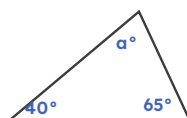


E2 Find a .



YOU DO Your turn.

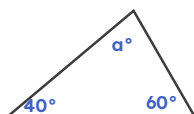
31 Find a .



$$a + 40 + 65 = 180$$

$$a = 180 - 105 = 75$$

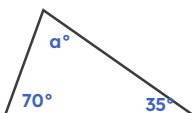
32 Find a .



$$a + 40 + 60 = 180$$

$$a = 180 - 100 = 80$$

33 Find a .



$$a + 70 + 35 = 180$$

$$a = 180 - 105 = 75$$

Challenging

Questions 34–39

Isosceles triangles have two equal angles (opposite the equal sides). Find each pronumeral.



WE DO With your teacher: copy the working.

E1 The apex angle is 24° . Find the base angles x .

E2 A base angle is 80° . Find the apex angle y .

YOU DO Your turn.

34 The apex angle is 52° . Find the base angles x .

$$2x + 52 = 180$$

$$2x = 128$$

$$x = 64$$

35 A base angle is 55° . Find the apex angle y .

$$y + 55 + 55 = 180$$

$$y = 180 - 110$$

$$y = 70$$

36 The apex angle is 104° . Find the base angles x .

37 A base angle is 70° . Find the apex angle y .

38 The apex angle is 116° . Find the base angles x .

39 A base angle is 65° . Find the apex angle y .

6.07 Exterior angle of a triangle

Easy

Questions 1–18

The exterior angle equals the sum of the two interior opposite angles.

WE DO With your teacher: copy the working.

- E1** interior opposite: 35° and 54° _____
- E2** interior opposite: 90° and 19° _____

YOU DO Your turn.

- 1** interior opposite: 86° and 56° 142°
- 2** interior opposite: 20° and 30° _____
- 3** interior opposite: 28° and 86° _____
- 4** interior opposite: 39° and 106° _____
- 5** interior opposite: 52° and 48° _____
- 6** interior opposite: 71° and 48° _____
- 7** interior opposite: 88° and 62° _____
- 8** interior opposite: 65° and 20° _____
- 9** interior opposite: 15° and 90° _____
- 10** interior opposite: 41° and 106° _____
- 11** interior opposite: 17° and 76° _____
- 12** interior opposite: 85° and 20° _____
- 13** interior opposite: 17° and 128° _____
- 14** interior opposite: 46° and 57° _____
- 15** interior opposite: 41° and 41° _____
- 16** interior opposite: 33° and 43° _____
- 17** interior opposite: 45° and 45° _____
- 18** interior opposite: 45° and 83° _____

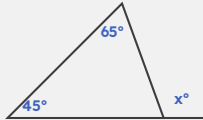
Medium

Questions 19–21

Find x , the exterior angle. Write the rule and the sum.

WE DO With your teacher: copy the working.

- E1** The interior opposite angles are 45° and 65° .



- E2** The interior opposite angles are 60° and 35° .



YOU DO Your turn.

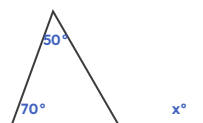
- 19** The interior opposite angles are 50° and 55° .



$$x = 50 + 55 \text{ (exterior angle)}$$

$$x = 105$$

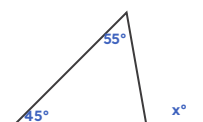
- 20** The interior opposite angles are 70° and 50° .



$$x = 70 + 50 \text{ (exterior angle)}$$

$$x = \underline{\hspace{2cm}}$$

- 21** The interior opposite angles are 45° and 55° .



$$x = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} \text{ (exterior angle)}$$

$$x = \underline{\hspace{2cm}}$$

Challenging

Questions 22–26

Use the exterior angle rule backwards to find the missing interior angle.

WE DO With your teacher: copy the working.

- E1** The exterior angle is 160° and one interior opposite angle is 65° . Find the other, y .
- _____
- _____

- E2** The exterior angle is 140° and one interior opposite angle is 95° . Find the other, y .
- _____
- _____

YOU DO Your turn.

- 22** The exterior angle is 120° and one interior opposite angle is 80° . Find the other, y .

$$y + 80 = 120$$

$$y = 120 - 80$$

$$y = \underline{\hspace{2cm}}$$

- 23** The exterior angle is 90° and one interior opposite angle is 55° . Find the other, y .

$$y + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

- 24** The exterior angle is 105° and one interior opposite angle is 25° . Find the other, y .
- _____
- _____

- 25** The exterior angle is 95° and one interior opposite angle is 70° . Find the other, y .
- _____
- _____

- 26** The exterior angle is 170° and one interior opposite angle is 135° . Find the other, y .
- _____
- _____

Easy

Questions 1–7

Name each quadrilateral from its description.

WE DO With your teacher: copy the working.

- E1** 4 equal sides and 4 right angles _____
- E2** 4 right angles, opposite sides equal _____

YOU DO Your turn.

- 1** 4 equal sides, no right angles rhombus

- 2** 2 pairs of parallel sides, no right angles _____

- 3** exactly 1 pair of parallel sides _____

- 4** 2 pairs of equal adjacent sides _____

- 5** a rectangle with 4 equal sides _____

- 6** a parallelogram with a right angle _____

- 7** a parallelogram with 4 equal sides _____

Medium

Questions 8–11

Name the quadrilateral, then give one property it has.

WE DO With your teacher: copy the working.

E1

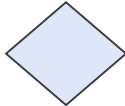


E2



YOU DO Your turn.

8



name: rhombus

property: all sides equal, opposite angles equal

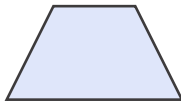
9



name: parallelogram

property: opposite sides are equal and parallel

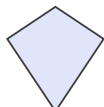
10



name: trapezium

property: one pair of parallel sides

11



Challenging

Questions 12–15

True or false? Give a reason.

WE DO With your teacher: copy the working.

- E1** Every square is a rectangle.
- _____

- E2** Every rectangle is a square.
- _____

YOU DO Your turn.

- 12** Every rhombus is a parallelogram.

true

- 13** A kite is a parallelogram.
- _____
- _____

- 14** Every square is a rhombus.
- _____
- _____

- 15** A trapezium has two pairs of parallel sides.
- _____
- _____

6.09 Angle sum of a quadrilateral

Easy

Questions 1–29

The angles of a quadrilateral add to 360° . Find the fourth angle.

WE DO With your teacher: copy the working.

E1 $60^\circ, 115^\circ, 105^\circ$ _____

E2 $90^\circ, 115^\circ, 115^\circ$ _____

YOU DO Your turn.

1 $115^\circ, 130^\circ, 75^\circ$ 40°

2 $55^\circ, 115^\circ, 115^\circ$ _____

3 $100^\circ, 120^\circ, 50^\circ$ _____

4 $105^\circ, 95^\circ, 75^\circ$ _____

5 $85^\circ, 65^\circ, 95^\circ$ _____

6 $50^\circ, 135^\circ, 120^\circ$ _____

7 $110^\circ, 80^\circ, 130^\circ$ _____

8 $120^\circ, 75^\circ, 120^\circ$ _____

9 $90^\circ, 85^\circ, 110^\circ$ _____

10 $110^\circ, 85^\circ, 75^\circ$ _____

11 $70^\circ, 70^\circ, 65^\circ$ _____

12 $100^\circ, 85^\circ, 135^\circ$ _____

13 $60^\circ, 95^\circ, 50^\circ$ _____

14 $70^\circ, 110^\circ, 85^\circ$ _____

15 $120^\circ, 85^\circ, 50^\circ$ _____

16 $130^\circ, 95^\circ, 85^\circ$ _____

17 $95^\circ, 50^\circ, 110^\circ$ _____

18 $95^\circ, 100^\circ, 50^\circ$ _____

19 $75^\circ, 70^\circ, 75^\circ$ _____

20 $130^\circ, 100^\circ, 95^\circ$ _____

21 $130^\circ, 50^\circ, 140^\circ$ _____

22 $110^\circ, 125^\circ, 85^\circ$ _____

23 $90^\circ, 110^\circ, 70^\circ$ _____

24 $50^\circ, 120^\circ, 135^\circ$ _____

25 $80^\circ, 125^\circ, 85^\circ$ _____

26 $130^\circ, 50^\circ, 125^\circ$ _____

27 $140^\circ, 50^\circ, 105^\circ$ _____

28 $130^\circ, 80^\circ, 60^\circ$ _____

29 $90^\circ, 50^\circ, 95^\circ$ _____

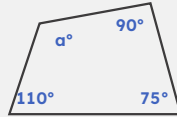
Medium

Questions 30–32

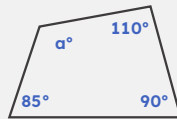
Find a . Write the equation first.

WE DO With your teacher: copy the working.

E1 Find a . (Not to scale.)

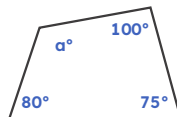


E2 Find a . (Not to scale.)



YOU DO Your turn.

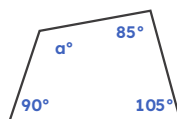
30 Find a . (Not to scale.)



$$a + 80 + 75 + 100 = 360$$

$$a = 360 - 255 = 105$$

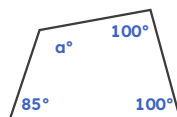
31 Find a . (Not to scale.)



$$a + 90 + 105 + 85 = 360$$

$$a = ______ - ______ = ______$$

32 Find a . (Not to scale.)



$$a + ______ + ______ + ______ = ______$$

$$a = ______ - ______ = ______$$

Challenging

Questions 33–36

Use the properties of the shape, then the angle sum of 360° .

WE DO With your teacher: copy the working.

E1 A parallelogram has an angle of 100° . Find the other three angles.

E2 A kite has two equal angles x° , and its other angles are 100° and 40° . Find x .

YOU DO Your turn.

33 A parallelogram has an angle of 150° . Find the other three angles.

opposite angle: 150°

co-interior: $180 - 150 = 30^\circ$

angles: $______^\circ, ______^\circ, ______^\circ, ______^\circ$

34 A kite has two equal angles x° , and its other angles are 110° and 70° . Find x .

$$______x + ______ + ______ = ______$$

$$______x = ______$$

$$x = ______$$

35 A parallelogram has an angle of 50° . Find the other three angles.

36 A kite has two equal angles x° , and its other angles are 100° and 100° . Find x .

Easy

Questions 1–10

True or false?

WE DO With your teacher: copy the working.

E1 The diagonals of a rectangle are equal. _____

E2 The diagonals of a rhombus meet at 90° . _____

YOU DO Your turn.

1 The diagonals of a parallelogram are equal. false

2 A square has equal diagonals that meet at 90° . _____

3 Opposite sides of a parallelogram are equal. _____

4 A kite has two pairs of equal opposite sides. _____

5 The diagonals of a parallelogram bisect each other. _____

6 A rhombus has equal diagonals. _____

7 The diagonals of a kite meet at 90° . _____

8 Every angle of a rectangle is 90° . _____

9 A trapezium always has a right angle. _____

10 Co-interior angles of a parallelogram add to 180° . _____

Medium

Questions 11–18

Use the properties to find the missing value.

WE DO With your teacher: copy the working.

E1 A rectangle has one diagonal 5 cm. How long is the other? _____

E2 A rhombus has a side of 9 cm. Find its perimeter. _____

YOU DO Your turn.

11 A parallelogram has an angle of 115° . Find the angle next to it.

co-interior angles add to 180°

$$180 - 115 = 65^\circ$$

12 A square has a perimeter of 32 cm. Find a side.

square: all sides equal

$$___ \div ___ = ___ \text{ cm}$$

13 A rhombus has an angle of 40° . Find the opposite angle.

rhombus: _____

$$___^\circ$$

14 A rectangle has one diagonal 7 cm. How long is the other? _____

15 A rhombus has a side of 6 cm. Find its perimeter. _____

16 A parallelogram has an angle of 120° . Find the angle next to it. _____

17 A square has a perimeter of 48 cm. Find a side. _____

18 A rhombus has an angle of 145° . Find the opposite angle. _____

Challenging

Questions 19–22

Give a reason why each statement is true.

WE DO With your teacher: copy the working.

E1 A square is a rectangle. _____

E2 A square is a rhombus. _____

YOU DO Your turn.

19 A rhombus is a parallelogram. because both pairs of opposite sides

20 A rectangle is a parallelogram. _____

21 A rhombus is a kite. _____

22 A square is a kite. _____